JPART 23:289–305

Expectations, Disconfirmation, and Citizen Satisfaction with the US Federal Government: Testing and Expanding the Model

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American Customer Satisfaction Index

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ABSTRACT

Recent research on citizen satisfaction with government services has examined the expectancy-disconfirmation model (EDM), a model suggesting that satisfaction judgments are formed through a cognitive process relating prior expectations to perceived performance and the confirmation or disconfirmation of expectations relative to performance. The results of these studies have been promising and largely supportive of the application of this model to the domain of government services, helping to clarify the processes by which citizens form satisfaction judgments about these services. Thus far, however, the model has only been tested in its most basic form and only applying survey data of citizens experiencing urban, local, or state government services. In this article, we expand on the extant research in two ways. First, we test the EDM in relation to US federal government services and compare our results to the findings of earlier studies focused on local government services. Second, we expand the model by including some antecedents we hypothesize will influence citizens' expectations of their experiences at the federal level of government, including the respondent's political ideology, party identification, and overall trust in the federal government. Our results suggest that the EDM functions well in regard to federal government services, confirming and building upon the findings of earlier studies of this model vis-à-vis local government services. Furthermore, analysis of the expanded EDM finds a significant relationship between party ID, ideology, trust and expectations, suggesting a new direction for future research using this model.

INTRODUCTION

While traditionally and primarily used to explore the formation of consumer satisfaction judgments with private sector goods and services (Oliver 1980), the expectancy-disconfirmation model (EDM) has recently been employed to examine the cognitive processes influencing citizen satisfaction with government service delivery (James 2009; Poister and Thomas 2011; Reisig and Chandek 2001; Roch and Poister 2006; Van Ryzin 2004, 2006, 2007;

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doi:10.1093/jopart/mus012
Advance Access publication April 19, 2012
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Van Ryzin et al. 2004). The EDM suggests a dynamic relationship between the prior expectations of consumers, post-experience perceptions of the quality (or performance) of the good or service, the confirmation or disconfirmation (either positive of negative) of these prior expectations based on perceptions of performance (the gap between expectations and reality), and, finally, resulting satisfaction with an experience. The first forays into the application of this model to the domain of government services have reported generally positive results, supporting its usefulness in efforts to clarify the sources of citizen satisfaction with government services (James 2009; Poister and Thomas 2011; Roch and Poister 2006; Van Ryzin 2004, 2006).

To date, however, the EDM has only been tested using survey data of citizens experiencing urban, local, or state government services, leaving open the question of the applicability of this model to other types of government services—most notably federal government services, where there is reason to believe that citizens' expectations may function quite differently. In this article, we expand on the extant research applying the EDM to government services in two ways. First, analyzing a sample of survey responses collected in 2010 from 1480 citizens who experienced a wide variety of US federal agencies and departments, we test the EDM at this level of government and compare the results to the findings of some earlier studies. Second, we build on the existing literature by expanding the model to include some important antecedents of citizen expectations likely to influence these expectations (especially at the federal level of government), including the respondent's political ideology, party identification, and general trust in federal government.

Our findings suggest that the EDM functions well in relation to citizens who experience federal government services, largely confirming the findings of earlier research. A comparison of the parameter estimates between this study and earlier studies where the model was applied to local government services, even taking into account slight differences in sampling, data collection methods, survey methodology, and statistical analysis, reveals a close relationship. As such, the utilization of this model toward a deeper understanding of how citizens form satisfaction judgments at the federal level of government and with federal services is warranted. Furthermore, analysis of our expanded EDM finds a significant relationship between party identification, ideology, trust, and expectations, suggesting a new direction for future research using this model and providing a deeper understanding of what determines expectations (and thus satisfaction) with federal government services.

The remainder of the article is structured as follows. In the next section (''The Expectancy-Disconfirmation Model''), we outline the EDM (or its extension, the ''expectancy disconfirmation with performance model''), discuss the extant literature applying this model to public sector services and make the case for both extending this research to include a unique test of citizen experiences with federal government services and for including additional antecedents of expectations to the model as it is applied to federal government services. In the section that follows (Data, Methods, and Findings), we describe the data we use to test the model, adopted from the American Customer Satisfaction Index (ACSI), and the methods used to collect the data. We then discuss the statistical techniques we use to analyze it and the results of our analysis (Statistical Methods and Findings), focusing on a comparison of these results to earlier research, as well as discussing the unique insights gained by including antecedents of expectations in the model. We conclude the article (Summary, Discussion, and Implications) with a discussion of how future research in this area might further expand knowledge of citizen satisfaction, the

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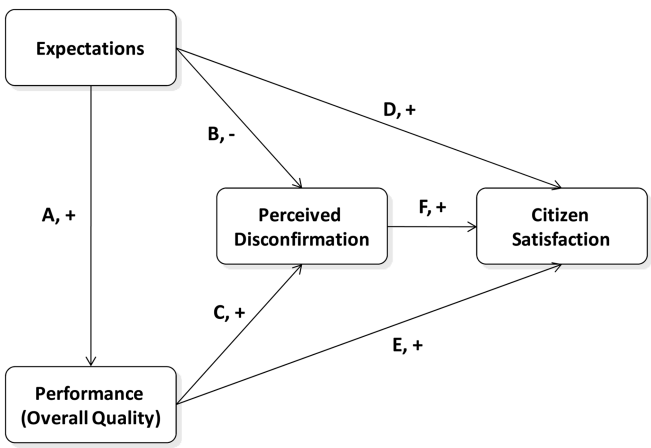
implications of our findings (to public managers, researchers, and others), and how government might use this information to improve both citizen perceptions of government performance and citizen satisfaction.

THE EXPECTANCY-DISCONFIRMATION MODEL

Theory, Model, and Applications to Government Services

The link between expectations, perceived quality (or perceived performance), the confirmation/disconfirmation of expectations, and satisfaction with a range of individual experiences—including life satisfaction, job satisfaction, consumer satisfaction, and so forth—has a long history in the academic literature (Campbell et al. 1976; Oliver 1980; Spector 1956). Pre-experience expectations, viewed as an initial referent against which actual experiences are compared during the formation of the individual's satisfaction judgments, have thus assumed a central role in research on satisfaction in these and related areas. Likewise, the interaction between theory building and model testing of the expectations–disconfirmation–satisfaction relationship has resulted in the refinement of this model over time (Anderson and Sullivan 1993; Oliver 1980, 1997).[1] In its most recent and most popular variant, the EDM takes the form represented in figure 1.

Figure 1
Expectancy Disconfirmation Model



-The "+/-" on each path (identified by letters, "A" through "F") reflects the hypothesized relationship between the variables in the model.

1 In what follows, what we refer to as the "EDM" is in fact a second-generation variant of this model, the expectancy-disconfirmation with performance model. However, given the popularity and wide use of the expectancy-disconfirmation with performance model—and even the conflation of the two models in some of the literature—we will focus only on the latter here.

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At its core, the EDM attempts to delineate the role of a subject's prior expectations in the cognitive processes through which satisfaction judgments regarding some (potentially any) experience are formed. Indeed, as an illustration of its diversity, in one of its first applications, a variation of the model was used by military psychologists to study the relationship between expectations of promotion within the ranks and resulting job satisfaction (Spector 1956). The central component of the model is the confirmation or disconfirmation of expectations (including both positive disconfirmation, implying better-than-expected performance, and negative disconfirmation, implying worse-than-expected performance) and resulting satisfaction with an experience. These cognitive processes are hypothesized to begin, obviously enough, with the individual's prior expectations regarding the experience in question.[2] As shown in figure 1, these prior expectations are thought to impact (both directly and indirectly) a range of consequent perceptions. First, in the aggregate, prior expectations of an experience should positively influence post-experience perceived performance (path "A" in figure 1). Expectations are formed from previous comparable experiences, first and foremost, but also through word-of-mouth, advertising, media influences, and so forth and because individuals are generally rational and often have long experience (even if only general experience) with the good or service they are experiencing, there should be a relatively small gap between what is expected and actual performance, where high expectations result in strong perceived performance and vice versa.

On the other hand, in the model prior expectations should negatively influence disconfirmation of expectations (path "B" in figure 1). That is, the individual's expectations can be high, middling (average), or low, and those with higher expectations are, *ceteris paribus*, more likely to have those expectations negatively disconfirmed (i.e., where performance fails to live-up to expectations). For similar reasons, those with low expectations are predicted to be more likely to have their expectations positively disconfirmed, whereas average expectations are more likely to be confirmed. Regarding performance, perceived performance should positively influence disconfirmation of expectations (path "C" in figure 1). That is, strong perceived performance will more likely lead to positive disconfirmation of expectations, all else being equal, whereas poor perceived performance will lead to negative disconfirmation of expectations.

Regarding the impact of these three variables on satisfaction, prior expectations are predicted to positively influence satisfaction (path "D" in figure 1). That is, because expectations form a baseline or "starting point" for satisfaction judgments (similar to their relationship to perceived performance), as expectations increase level of satisfaction is predicted to increase as well. In the EDM, positive/negative disconfirmation takes satisfaction away from the baseline provided by expectations. Thus, disconfirmation of expectations is predicted to positively influence satisfaction (path "F" in figure 1). Positive disconfirmation of expectations drives satisfaction higher from the baseline level set by prior expectations, whereas negative disconfirmation does the opposite. Finally, performance is predicted to positively influence satisfaction (path "E" in figure 1). That is, post-experience perceptions of actual performance matter as much (and typically more) than either prior

2 It should be noted that throughout the article, we rely on a "rational" or empirical definition of expectations rather than a "normative" one. Regarding the latter view, expectations are framed as what the respondent expects "should" (ideally) be forthcoming through some experience. On the other hand, for a rational or empirical definition of expectations, as operationalized in the survey questionnaire and data we analyze below, the respondent is simply asked what they expected to receive based on their knowledge, prior experiences, and so forth.

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expectations or disconfirmation of expectations in determining an individual's level of satisfaction.

As mentioned earlier, a large number of mostly academic studies have tested and confirmed the EDM in relation to a range of individual experiences, including (and predominantly) the processes leading to consumer satisfaction with a variety of products and services (Churchill and Surprenant 1982; Cadotte, Woodruff, and Jenkins 1987; Fornell et al. 1996; Oliver 1980). Yet considering the long history of research into this model as a key determinant of consumption satisfaction across a variety of contexts, the field of public administration is relatively late to the game in utilizing it to explain citizen satisfaction with government services. Nevertheless, over the last few years this model has drawn increased attention among those interested in understanding the cognitive processes responsible for citizen satisfaction and dissatisfaction with government services, with a modest but growing body of literature now available (James 2009; Poister and Thomas 2011; Reisig and Chandek 2001; Roch and Poister 2006; Van Ryzin 2004, 2006, 2007; Van Ryzin et al. 2004).

A handful of these government-specific studies have tested the EDM represented above directly (Van Ryzin 2004, 2006) and have predominantly confirmed the hypothesized relationships included therein. Others have looked more generally at the role of expectations and/or disconfirmation in the citizen satisfaction formation process without testing the EDM directly but still find results consistent with a role for expectations in explaining satisfaction (James 2009; Poister and Thomas 2011; Reisig and Chandek 2001; Roch and Poister 2006). Furthermore, some of these studies have investigated the model as it relates to a specific local government service like policing (James 2009; Reisig and Chandek 2001), others have examined it across a broad range of local/urban services (Van Ryzin 2004, 2006), and still others have examined expectations among state government–level respondents (or state government–specific services) (Poister and Thomas 2011; Roch and Poister 2006).

Because the development of a body of scholarly literature examining the EDM in the context of government services is a new and on-going process, there remain several unanswered questions about this model's application to the domain of government services. In the first instance, the model has not yet been tested with survey data regarding all levels of government—more specifically, none of the aforementioned studies have tested the model vis-à-vis the federal or national level of government, leaving its applicability to this level in doubt. Without such an empirical test, we have little reason to anticipate how this model will perform when applied to federal government services, and its empirical performance could prove contrary to expectations. Indeed, there are strong reasons to believe that citizen expectations regarding government services at the federal level in the United States may function significantly differently than at other levels of government (urban/local or state), and therefore that the EDM itself may function significantly differently.

Consider for a moment some of the reasons why citizen's expectations with federal government services may differ from their expectations with local or state government services, differences that could result in distinct relationships within the EDM. In the first instance, the types of services citizens receive from state and local governments often differ significantly from the types of services they experience with the federal government. The studies discussed earlier focus mostly on citizen perceptions of services like waste disposal, policing, fire protection, and so forth, whereas the "average" citizen experiences vastly

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different services when encountering the federal government—such as benefits distributed by the Social Security Administration, taxes collected by the Internal Revenue Service, and so forth. Because the nature of these services differ so dramatically, it is reasonable to infer that expectations, and the relationships between expectations, perceived performance, disconfirmation, and so forth, may differ as well. For example, the significantly greater importance of applying for Social Security benefits or filing federal taxes when compared to having one's garbage collected (at least for most citizens) likely means that citizens have given far greater thought to the first two processes, thought that could impact both how expectations are formed and how those expectations impact perceived performance, confirmation/disconfirmation, and satisfaction.

Perhaps more importantly, however, is the way that citizen's perceptions of the federal government as a whole may influence their expectations with services delivered by this level of government. Studies have consistently shown that citizens tend to trust state and local government far more than the federal government, due possibly to the greater "closeness" of these local institutions to citizens' everyday lives when compared to more "distant" federal institutions (CNN 2010; Gallup 2008). Interestingly, as trust in the federal government has declined over the past few decades, trust in state and local governments has trended in the other direction (Hetherington and Nugent 2001). The far higher (and improving) levels of overall trust in state and local government institutions raises the possibility—and in fact, makes it likely—that citizens enter their experiences with federal government services with different (i.e., lower) expectations, whereas possibly also changing the way expectations relates to the other variables in the EDM.

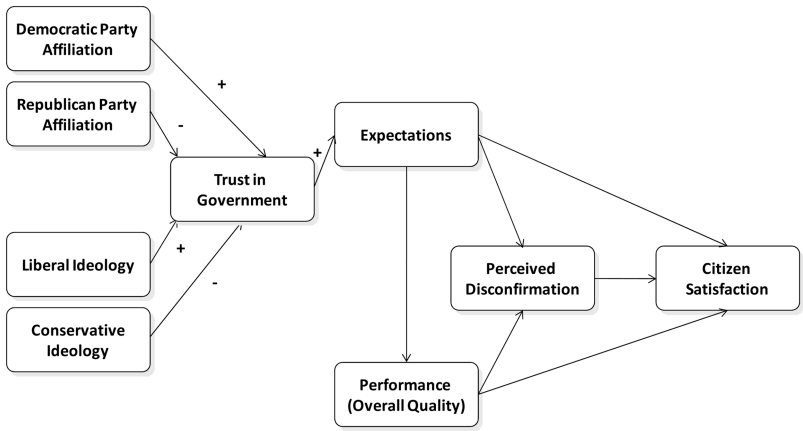
Moreover, when compared to state and local government, citizen perceptions of the federal government tend to be influenced to a larger degree by pre-existing political dispositions like party identification and political ideology. Thus, while party ID and ideology are usually found to have little impact on perceptions (such as trust) in state or local governments, they have been shown to significantly impact trust in the federal government, with those on the right side of the political spectrum (in both party ID and ideology) less likely to trust the federal government than others, for example (Uslaner 2001). As such, it is reasonable to suspect that party ID and ideology are also likely to drive citizen expectations with federal government services, a fact which could also impact the results derived from the EDM.

Taken together, the facts outlined above suggest the need for testing the EDM at the federal level of government, where it has not yet been tested, but they also suggest a path for expanding the model that could prove particularly useful when applying it to this level of government. We propose, therefore, to examine an extended version of the EDM for the federal government, represented below in figure 2.

In this modified version of the EDM, the relationships within the basic model remain intact, yet we hypothesize that certain pre-existing perceptions of the federal government will significantly impact citizens' prior expectations with federal services. In the first instance, as mentioned above, we hypothesize that overall, general trust in the federal government is likely to influence citizens' prior expectations with the services they receive, with those expressing greater overall trust in the federal government also holding higher expectations of the services they will receive and vice versa. Furthermore, and consistent with the prior literature mentioned above, we hypothesize that political ideology and party ID will significantly impact overall trust in the federal government and thereby (indirectly)

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Figure 2
Expectancy Disconfirmation Model with Antecedents of Expectations



impact prior expectations as well. Consistent with the findings of these earlier studies, we would expect that those identifying themselves as "Liberal" or a "Democrat" will exhibit greater trust in the federal government, whereas those identifying themselves as "Conservative" or a "Republican" will exhibit lower trust. And, assuming a significant relationship between trust and expectations, these factors will therefore influence prior expectations as well.

DATA AND STATISTICAL METHODS

Data

The data we use to test the EDM and the expanded version of this model outlined above come from the ACSI (Fornell et al. 1996). An established national indicator of consumer satisfaction, the ACSI annually measures satisfaction (as well as an assortment of antecedents and outcomes of satisfaction) for a wide range of private sector companies and industries across all the major economic sectors. Since 1994, ACSI has also measured citizen satisfaction with local, state, and federal government services. In the area of citizen satisfaction with government services, ACSI data have been used to compare satisfaction across the public and private sectors (Morgeson and Mithas 2009), to examine the nature and determinants of citizen satisfaction with federal e-government (Morgeson 2011; Morgeson, VanAmburg, and Mithas 2011), and to compare determinants and outcomes of satisfaction across various US federal agencies and departments (Morgeson and Petrescu 2011).

The ACSI federal government data set used in this study comes from the 2010 wave of the project, collected during the months of July and August of 2010. The 2010 data set includes a total of 1480 respondents, all of whom indicated having some interaction with a federal government department or agency in the past year (excluding those respondents who experienced only the US Postal Service). Two sampling and interviewing methods

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were used to collect this data. The first method used was computer-assisted telephone interviewing (CATI), with trained interviewers working from a computer terminal containing the questionnaire script and interviewing respondents from this script over the telephone. Random-digit-dial probability sampling (where telephone numbers are dialed randomly from within area codes based proportionally on the population included within geographic areas) and multiple call-back and refusal conversion techniques (minimizing the impact of nonresponse bias) were also implemented. Using these methods, slightly less than half of the sample ($n = 696$) was collected.

The other roughly half of the sample ($n = 784$) was collected using online panel/web-based interviewing methods, a method that differs slightly from the CATI-RDD methods described above. Employing a large panel of email addresses (corrected to Census demographics) and selecting randomly from this panel, invitations are sent by email to potential respondents (along with multiple reminder emails) soliciting participation in the study. Those choosing to participate via this method were directed to a secure website programmed to administer the questionnaire, with the respondent effectively self-administering the formatted survey.[3]

Via both interviewing methods, potential respondents to the questionnaire were screened prior to the start of interviewing for recent personal experience with a federal agency before being determined eligible to participate. Only respondents indicating actual contact with a federal agency (again, excluding the US Postal Service) over the prior 12 months were eligible to participate. This screening method ensures that respondents have perceptions based on actual experiences with a federal agency (a feature vital to accurately measuring perceived performance, disconfirmation, and satisfaction), rather than having only abstract opinions of federal services. Across the sample, interviewees identified a total of 57 distinct federal programs, agencies, or departments experienced.[4] Those qualifying for and choosing to participate in the study were asked a range of questions regarding their experiences with the agency, including questions about their expectations, the quality of the services received, their satisfaction, and so forth.

Table 1 provides questions and question wording for the variables from the ACSI data set focused on in this study. Table 2 provides descriptive statistics for all these variables as well.

3 The 784 cases collected online were collected using the Research Now web panel, a non-probability, double opt-in panel of respondents interviewed via the Internet. While a sample drawn from this type of panel can differ from a probability sample (like the 696 respondents identified and interviewed through RDD/CATI), prior to our analysis the two portions of the sample were examined for demographic differences, as well as significant differences in the variables themselves, potentially caused by these distinct sampling and interviewing methods. No noteworthy or potentially confounding differences were discovered.

4 Respondent heterogeneity within the sample, in relation to the diverse assortment of federal agencies and departments experienced by respondents to the survey, and its impact on the model estimates should be noted here. That is, there is no reason to believe, nor are we suggesting, that the EDM functions exactly the same across these various subgroups. In fact, a cursory test (for which the results are not shown) indicates that the EDM estimates do indeed exhibit significant differences across many of these respondent subgroups. However, because of limitations in sample size for some of these groups, as well as what we suspect would be limited general interest in a distinct test of this model across dozens of federal agencies, the model will not be reestimated separately for each agency subgroup within the sample.

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Table 1
Questions and Question Wording (Abbr.) in ACSI Sample

Expectations	How would you rate your expectations of the overall quality of services from the (AGENCY/DEPARTMENT)?
Overall quality	How would you rate the overall quality of the (AGENCY/DEPARTMENT)'s services?
Confirmation of expectations	Considering all of your expectations, to what extent have the (AGENCY/DEPARTMENT)'s services fallen short of your expectations or exceeded your expectations?
Overall satisfaction	First, please consider all your experiences to date with the (AGENCY/DEPARTMENT)'s services. How satisfied are you with the (AGENCY/DEPARTMENT)'s services?
Trust	Generally speaking, how much of the time do you think you can trust the government in Washington?
Ideology	We hear a lot of talk these days about liberals and conservatives. When it comes to politics, do you usually think of yourself as extremely liberal, liberal, slightly liberal, moderate or middle of the road, slightly conservative, extremely conservative, or haven't you thought much about this?
Party identification	Generally speaking, do you usually think of yourself as a Republican, a Democrat, an Independent, or something else? (IF REPUBLICAN OR DEMOCRAT) Would you call yourself a strong (REP/DEM) or a not very strong (REP/DEM)?

Statistical Methods and Findings

To estimate the coefficients within both the EDM and the expanded version of the model outlined earlier, we utilize full information maximum likelihood estimation (FIML) and estimate structural equation models (Arbuckle 2006). An accepted and well-used procedure for estimating structural models, FIML estimates multiple maximum likelihood equations simultaneously and includes information from the conditional covariance among all model variables into the model parameter estimates. The FIML technique has been shown to outperform least squares regression in a variety of contexts (Enders 2001; Enders and Bandalos 2001) and has been used in other similar studies when applying the EDM to government services (Van Ryzin

Table 2
Descriptive Statistics

	N	Minimum	Maximum	Mean	Standard Deviation
Expectations about overall quality	1450	1	10	7.152	2.507
Perception of overall quality	1466	1	10	7.559	2.410
Confirmation/disconfirmation of prior expectations	1459	1	10	6.727	2.548
Overall citizen satisfaction	1466	1	10	7.381	2.507
Democratic party ID dummy (Strong Democrat = 1)	1480	0	1	0.218	0.413
Republican party ID dummy (Strong Republican = 1)	1480	0	1	0.147	0.355
Extremely liberal ideology dummy (Extremely Liberal = 1)	1480	0	1	0.051	0.219
Extremely conservative ideology dummy (Extremely Conservative = 1)	1480	0	1	0.064	0.244
Trust in the government in Washington, DC	1441	1	10	4.670	2.628

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2004, 2006), all facts recommending its use in our study. The zero-order correlation matrices for all variables included in our structural models are provided in table 3.

Results for the structural model for the EDM, including standardized direct effects, significance of the coefficients, and measures of explained variance (squared multiple correlations), are presented below in Model 1. A comparison of the results from Model 1 to those from an earlier study is provided in table 4. A discussion of these findings follows.

Taken together, the results shown in Model 1 largely support the application of the EDM to the domain of federal government services, with the model performing mostly as expected. The observed results also largely confirm the findings of earlier studies that examine the EDM at other levels of government (i.e., for local government). Looking at the parameter estimates more closely, the relationship between prior expectations and perceived performance ($\beta = 0.52$, $p < 0.001$) is significant and in the expected direction, suggesting that for citizens experiencing federal government services, higher expectations do in fact frame and lead to stronger perceptions of experienced performance. The coefficient is also reasonably similar to the findings from Van Ryzin's (2006) study shown in table 4 ($\beta = 0.52$ versus $\beta = 0.35$), although expectations seem to more strongly influence perceived performance with federal services when compared to local services.

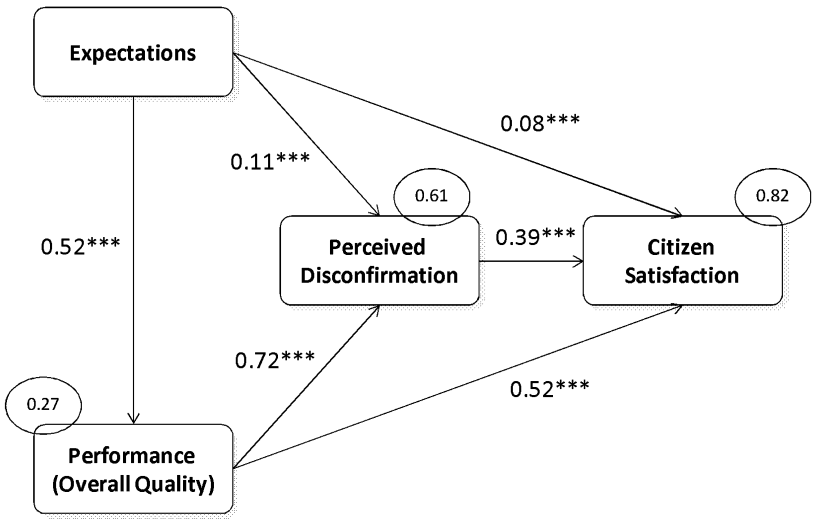
Table 3
Zero-Order Correlations

	1	2	3	4	5	6	7	8
1. Expectations about overall quality	—	—	—	—	—	—	—	—
2. Perception of overall quality	0.524**	—	—	—	—	—	—	—
3. Confirmation/ disconfirmation of prior expectations	0.487**	0.776**	—	—	—	—	—	—
4. Overall citizen satisfaction	0.543**	0.863**	0.832**	—	—	—	—	—
5. Democratic party ID dummy (Strong Democrat = 1)	0.070**	0.108**	0.115**	0.117**	—	—	—	—
6. Republican party ID dummy (Strong Republican = 1)	−0.064*	−0.044	−0.050	−0.045	−0.219**	—	—	—
7. Extremely liberal ideology dummy (Extremely Liberal = 1)	0.018	−0.001	0.012	−0.013	0.259**	−0.061*	—	—
8. Extremely conservative ideology dummy (Extremely Conservative = 1)	−0.060*	−0.046	−0.050	−0.053*	−0.117**	0.330**	−0.060*	—
9. Trust in the government in Washington, DC	0.321**	0.341**	0.374**	0.360**	0.336**	−0.198**	0.125**	−0.208**

*Significant at $p < .05$, **significant at $p < .01$.

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Model 1



Standardized estimates along model paths; ***Significant at p<.001; **Significant at p<.01; *Significant at p<.05
Squared multiple R's for each endogenous variable included in circles attached to variables

Note: Typically-reported model fit statistics could not be calculated for this model because the estimated model has zero degrees of freedom (i.e. the model is "saturated", or has an equal number of matrix covariances as estimated parameters).

The relationship between expectations and perceived disconfirmation ($\beta = 0.11$, $p < 0.001$) is the one somewhat unexpected result apparent in the model and thus deserves particular emphasis. Although the standardized coefficient is significant, it is not in the direction hypothesized by the EDM. Indeed, for citizens experiencing federal government

Table 4
Comparison to Earlier Studies[a]

	Morgeson (2012)	Van Ryzin (2006)
Expectations → Performance	0.52	0.35
Expectations → Disconfirmation	0.11	0.03
Performance → Disconfirmation	0.72	0.68
Expectations → Satisfaction	0.08	0.10
Performance → Satisfaction	0.52	0.41
Disconfirmation → Satisfaction	0.39	0.49

[a]Both sets of results reflect standardized parameter estimates computed using FIML. Two differences between how the VanRyzin results were modeled and our own approach are worth noting. First, VanRyzin modeled the "Expectations to Performance" link as a correlation rather than a direct effect (or so it appears, given that he presents a "curved path" in the structural model). But since Expectations are an exogenous variable in the model and are the only variable to influence Performance, this difference should have little consequence. Second, Van Ryzin's Performance measure is a latent variable containing several observed variables relating to local government performance, whereas we use an "overall quality" variable as our measure of Performance. Regardless of these different approaches, the results between the two studies are largely consistent.

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services, stronger expectations tend to result in a greater probability of *positive* disconfirmation. While unexpected and contrary to theory, this result is reasonably consistent with the findings from the Van Ryzin study ($\beta = 0.11$ versus $\beta = 0.03$), which suggests that stronger expectations do not appear to result in a greater probability of negative disconfirmation of expectations with government in general. This seemingly persistent finding of a null or even positive relationship between expectations and perceived disconfirmation deserves greater attention and theoretical development in the public administration literature. But it is likely explained by the fact that citizens tend to have unrealistically depressed expectations of the services they'll receive from government—and from federal government especially, given the widespread and well-publicized criticism of this level of government from all quarters. In turn, these unusually depressed expectations result in positive disconfirmation among most citizens, with services exceeding expectations once actually experienced, than the model predicts.

On the other hand, the relationship between performance and perceived disconfirmation is confirmed, with performance strongly, positively and significantly related to disconfirmation ($\beta = 0.72$, $p < 0.001$). In this case, the federal government results very closely match the findings from the local government context ($\beta = 0.72$ versus $\beta = 0.68$), with nearly identical parameter estimates observed across the two contexts.

Looking lastly at the three variables that directly impact citizen satisfaction in the EDM, expectations are found to be positively and significantly related to satisfaction ($\beta = 0.08$, $p < 0.001$) as anticipated, although the effect is fairly small. Here too the findings for the federal government match those from the model as applied to the local government context ($\beta = 0.08$ versus $\beta = 0.10$). Perceived disconfirmation also positively and significantly impacts satisfaction ($\beta = 0.39$, $p < 0.001$), indicating that citizens who have their prior expectations of federal services exceeded are also more satisfied with their experiences, as hypothesized within the model. In this instance, disconfirmation of expectations are not as strong a determinant of satisfaction in the federal government model as they are in the local government context ($\beta = 0.39$ versus $\beta = 0.49$). Finally, the relationship between performance and satisfaction is positive and significant ($\beta = 0.52$, $p < 0.001$), as hypothesized in the EDM, suggesting that stronger performance (or better perceived service quality) does in fact lead to greater satisfaction. Unlike the disconfirmation-satisfaction relationship, however, in this case the performance-satisfaction relationship is stronger in the federal government test of the model than in the local government test ($\beta = 0.52$ versus $\beta = 0.41$).

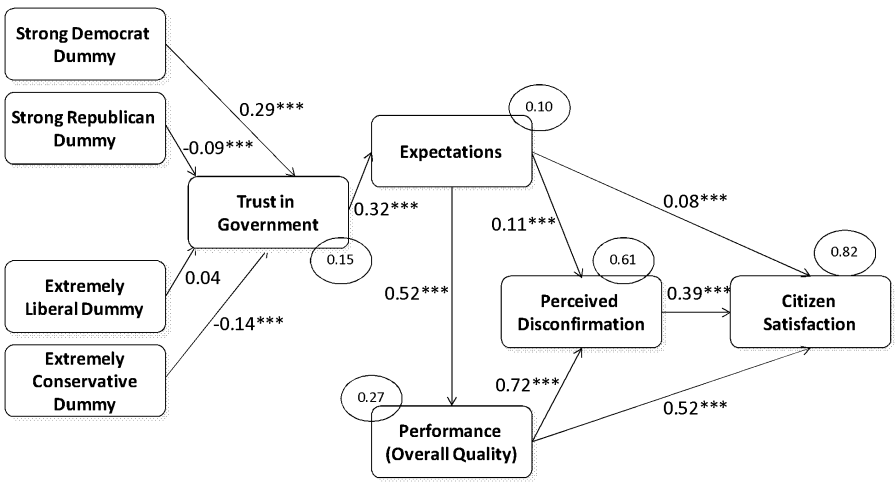
In sum, for our test of the EDM in relation to federal government services, and when compared to earlier tests at the local level of government, two sets of relationships stand out most, in our opinion. First, for the federal government, the relationships between expectations and performance and from performance to satisfaction are both stronger than in the local government context. That is, prior expectations with the federal government will more strongly impact perceptions of performance, which in turn will more strongly influence citizen satisfaction, suggesting a greater role for prior expectations in driving satisfaction at the federal level of government (especially considering that the relationships from expectations to disconfirmation and satisfaction are relatively small and roughly equal across the two models). Second, the relationship between disconfirmation and satisfaction is stronger in the EDM applied to local government services when compared to the federal government results. In short, this finding suggests that "positive surprises" at the local level

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will have a stronger impact on satisfaction at this level than at the federal level. We will discuss the practical implications of these and other important findings in greater detail in the concluding section of the article.

Beyond providing a first test of the EDM vis-à-vis federal government services and adding to the existing literature in this way, we also test an expanded version of the model that includes some antecedents of expectations with federal services, as discussed earlier. Results for this test are presented in Model 2 below.

Model 2



Standardized estimates along model paths; ***Significant at p<.001; **Significant at p<.01; *Significant at p<.05
Squared multiple R's for each endogenous variable included in circles attached to variables
Model Fit Statistics: $X^2(19) = 120.87$, P = 0.00; $X^2/df = 6.36$; RMSEA = 0.60

The results provided in Model 2 for the expanded EDM are predominantly consistent with what was anticipated.[5] Looking only at the additional variables added to the left-hand side of the model (and not the core model itself, as it is unchanged), we find that trust in government does in fact positively and significantly impact prior expectations ($\beta = 0.32$, $p < 0.001$). This one variable alone explains roughly 10% of the variance in expectations ($R^2 = 0.10$), suggesting that citizens' trust in the federal government moderately influences their expectations of what they will receive from the federal government. Moreover, three

5 Through preliminary tests of this expanded model, we found that the four hypothesized predictors of trust—the two political party variables and the two ideology variables—performed better when dichotomized and limited to the "extreme" category respondents (i.e., those respondents identifying themselves as "a Strong Republican," "Extremely Liberal," etc.), explaining why only this smaller subset of respondents were included as predictors of trust. For this reason, and also to compensate for the fact that some consider it inappropriate to treat categorical, Likert-type scale variables as continuous predictor variables in structural equation modeling, the variables were dichotomized prior to inclusion in the model produced here.

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of the four predictors of trust in government added to the model are significant and in the expected direction. Both those identifying themselves as "strong Republicans" ($\beta = -0.09$, $p < 0.001$) and those identifying themselves as "extremely conservative" ($\beta = -0.14$, $p < 0.001$) are less trusting of the federal government and, by extension, have deflated expectations of the services they will receive from the federal government. Conversely, those who identify themselves as "strong Democrats" ($\beta = 0.29$, $p < 0.001$) have greater trust in the federal government and thus come to their service experience with higher expectations. The relationship between those who self-identify as "extremely liberal" and trust is insignificant.

SUMMARY, DISCUSSION, AND IMPLICATIONS

In this study, we have tested the EDM by applying it to the domain of federal government services and thereby filling a gap in the existing literature, where tests have thus far focused only on the local or state government levels. We have also tested an expanded version of this model that includes determinants of citizen expectations with federal services. Our findings suggest that the standard EDM functions well vis-à-vis federal government, with observed parameters very similar to those found in earlier studies examining other levels of government. The expanded EDM also shows promise in relation to federal government, beginning a process of identifying those factors responsible for driving prior expectations with these services. Taken together, these findings suggest that the EDM can help clarify the cognitive processes by which citizens form satisfaction judgments about their experiences with federal government services, and thus help researchers, public managers, and others better understand citizen satisfaction with this level of government.

Before proceeding to further discuss the implications of our findings, some limitations of this study, and suggestions for how these limitations can be overcome in future research, should be mentioned. The results presented here are based on a single sample of citizens who interacted with a federal government program, agency, or department over a defined and relatively narrow period of time, and we must therefore be somewhat cautious in interpreting these results. This is particularly true in this case, given that we are dealing with a federal government which, it is safe to say, is currently an even more politicized environment than normal (at least at the time of this writing). As we have argued above, however, political ideology, party identification, and thus the relationship between these factors, and the political leanings of the current administration can substantially impact the perceptions of citizens, including their trust in government and their expectations of the services provided. As such, testing these models over time and across political administrations would be useful, helping to identify the stability of the EDM and the expanded EDM under changing political conditions.

Nevertheless, several important conclusions with significant implications—some methodological, and others practical—can be gleaned from this study. In the first instance, this study provides guidance to future researchers investigating the EDM vis-à-vis government services, in where the model can be confidently applied (i.e., at the federal level), how it can be expected to differ across levels of government (i.e., federal versus local government applications) and in how it might be developed further in the future (i.e., by expanding the model with antecedents of expectations). Our study confirms that the EDM works at the federal level and for citizens experiencing federal services and that the model provides relatively consistent results across different levels of government, with a few exceptions.

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Perhaps most importantly, our study has provided guidance on how the EDM can be expanded to provide even further insights into the role of antecedents of expectations in citizen satisfaction formation. These kinds of extensions are particularly important when considering this model's application to the domain of federal services, where political dimensions (trust, party, ideology) are likely to be more relevant in driving expectations than at other levels of government.

Regarding the more general practical implications of our study, these findings can certainly help public managers at the federal level better understand how their agencies or programs are faring in satisfying the citizens they come into contact with. Although manipulation of expectations is often more difficult for financially limited public sector organizations, and possibly even contrary to the objectives of an open, non-coercive government, these results can certainly inform managers about certain disadvantages they may face in achieving the goal of citizen satisfaction. For example, the knowledge that declining trust will create an environment of deflated expectations that in turn may prove detrimental to citizen satisfaction can certainly prepare public managers for a more challenging service delivery environment. Moreover, the knowledge that an agency's group of citizen–customers trend politically one way or the other can help managers understand the source of the higher (or lower) expectations citizens bring to their interactions with the agency.

Furthermore, while requiring creative interpretation of the empirical results and thus more open to competing explanations, some of the important and/or surprising results we highlighted earlier could themselves have significant practical implications for federal agencies, public managers within those agencies, and so forth. For example, what might we learn from the finding that expectations play a larger role in determining satisfaction at the federal level (indirectly, through the larger impacts of expectations on perceived performance and of performance on satisfaction noted earlier) when compared to the local level? The explanation for this finding may be as simple as the greater importance or more intensive nature of the experiences (i.e., filing federal taxes or applying for SSA benefits) citizens have with federal government, which might in turn condition them to hold more rational, realistic, and carefully considered expectations of their experiences. Yet because this finding suggests that expectations have a more potent framing effect for federal services when compared with local services—that is, that citizens experiencing federal services are more likely to "get what they think they'll get"—it indicates that managing expectations is even more important in this domain than in others. It also points to a difficult dilemma for the federal government. Because of the aforementioned financial (and possibly ethical) limitations on active government manipulation of expectations, periods of declining expectations caused by either internal or external (political) forces will likely be difficult to overcome and will therefore likely have a detrimental effect on satisfaction that agencies have little control over.

Related to this, but pointing to a positive role for government in improving satisfaction with federal services, how do we interpret the finding that citizens' perceptions of performance more powerfully determine satisfaction at the federal level when compared to the local level? Put differently, what is it about the nature of federal services that agency performance in the delivery of these services is more influential in driving satisfaction? While it is difficult to pinpoint the precise explanation for this finding without additional research—although it may be as simple as citizens believing that the services offered by the federal government themselves are more vital and thus agencies' performance in providing these services proves more important in satisfying citizens—the central thrust of this finding is clear. For federal

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agencies, it is even more critical that they focus on tangible quality improvements in seeking happier, more satisfied citizen–customers. In turn, and contrary to the less controllable impact of shifting expectations mentioned above, in this instance, federal agencies are at minimum given a clear and somewhat controllable path to meaningful satisfaction gains via internal performance and service quality improvement initiatives.

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